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A Hybrid Multi-Model Framework for Personalized User-Level Anomaly Detection with Data-Driven Threshold Optimization

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Abstract

Modern user authentication systems increasingly need user and device-behavior-aware adaptive mechanisms to detect evolving threats beyond the traditional authentication framework of static credential verification. This paper proposes a hybrid multi-model framework for personalized user-level anomaly detection using a data-driven Hybrid Anomaly Score (HAS). The primary contribution lies in deriving the HAS using the joint integration of three adaptive attributes: dynamically computed per-user deviation thresholds conditioned on individual behavioral history, profile-age-aware baseline weights reflecting user cohort maturity, and criticality-scaled aggregation with the security impact of each detection methodology. The framework is evaluated on a large-scale real-world dataset and demonstrates strong detection performance, while achieving low inference latency suitable for real-time enterprise deployment. The ablation analysis of the framework confirms that dynamic weighting and personalized threshold substantially improve detection stability and convergence with an effective and deployable solution for large-scale authentication environments.

Keywords: user authentication; anomaly detection; hybrid anomaly detection; cybersecurity; multi-factor authentication (MFA)

1. Introduction

Modern authentication systems are moving beyond the static Multi-Factor Authentication (MFA) to adaptive and risk-based mechanisms to balance security and usability. Risk-Based Authentication (RBA) systems dynamically assess contextual properties of user and device, including location and behavioral patterns to dynamically adjust authentication requirements [1–4]. The core of the RBA systems is user-level anomaly detection, which determine the deviation of the user authentication events from the legitimate historical behavior [5,6]. Despite the evolving progress, existing anomaly detection approaches have three fundamental limitations. First, many anomaly systems rely on static global thresholds, resulting in high false positive rates for new users of an authentication system. Second, several approaches use either single anomaly detection methods or loosely coupled multi-model frameworks, making it difficult to consolidate heterogeneous anomalies. Third, existing anomaly evaluations emphasize aggregate performance metrics, thus lacking contributions of individual design components and limiting the ability to assess generalization beyond the evaluated data set.



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Recent advances in adaptive authentication and risk-based access control have emphasized the importance of incorporating behavioral, contextual, and device-level signals to enhance security beyond traditional credential-based mechanisms [1–4,7]. In particular, anomaly detection techniques have been widely adopted to model user-specific behavioral baselines and detect deviations indicative of potential compromise [5,6]. However, the effectiveness of such systems depends heavily on the ability to integrate heterogeneous signals while maintaining scalability and low latency in enterprise environments [8,9].

Under user-level anomaly detection, deviations are evaluated relative to dynamically evolving behavioral baselines. We hypothesize that consolidating (i) per-user adaptive deviation thresholds, (ii) profile-age-aware weighting of different anomaly detectors, and (iii) criticality-sensitive aggregation of different anomaly detectors can significantly improve the reliability of the anomaly detection while maintaining real-time performance of the system.

To evaluate these hypotheses in practice, we propose a hybrid multi-model anomaly detection framework that integrates six different anomaly detection methodologies spanning user/device behavior, user authentication failures, user login patterns, geographic behavior, IP reputation, and device and mouse fingerprinting. The framework introduces a metric, i.e., unified Hybrid Anomaly Score (HAS) that aggregates output of each anomaly detector using analytically derived weights informed by anomaly rates, user profile maturity, and detector criticality.

This work is evaluated under a realistic enterprise-level threat model that includes credential abuse, automated attacks, and partial trust compromise. Such threat scenarios are consistent with modern authentication threat landscapes and Zero Trust security principles, where continuous verification and contextual risk assessment are essential [1,2,10,11]. The detailed threat model is defined in Section 3.1.

To address the limitations of existing anomaly detection approaches and to systematically evaluate the effectiveness of the proposed hybrid framework, it is essential to formulate clear research questions that guide both the design and evaluation of the system. These are the important research questions for this paper that would consider the important aspects like personalized behavior, profiling awareness, user behavior and real-time performance. Based on the above research questions, hypotheses have been proposed for further empirical testing.

The upcoming subsections present the key research questions, hypotheses, and contributions of the study to systematically guide the design and evaluation of the proposed framework.

1.1. Research Questions and Hypotheses

Some of the compelling research questions for the research are the following:

RQ1: How can the use of per-user deviation thresholds improve anomaly detection as compared to using static global thresholds?

RQ2: What is the influence of the maturity of the user profile (age of user profile) on anomaly rate and stability of detection in a large-scale authentication environment?

RQ3: Is there any kind of convergence of aggregation of user profile age and respective anomaly rate?

RQ4: Does the proposed hybrid anomaly detection framework achieve millisecond-level performance for per-authentication request under normal operational loads?

Based on these research questions, the following hypotheses are evaluated in practice:

Hypothesis 1 (Personalized Threshold Hypothesis). *Dynamically computed per-user deviation thresholds significantly reduce false-positive rates compared to existing static thresholding approaches.*

Hypothesis 2 (Profile Maturity Hypothesis). *User profile-age-aware weighting improves anomaly detection stability by reducing false positives for established users while maintaining sensitivity for new users.*

Hypothesis 3 (Weight Stabilization Hypothesis). *The more historical the data, the more stable the dynamic weights based on anomaly rates.*

Hypothesis 4 (Real-Time Operational Latency Hypothesis). *The proposed framework works well in real-time scenarios, with measured latency in milliseconds.*

1.2. Contributions

The proposed work uses multiple models to adaptively determine the anomaly. The framework combines multiple different anomaly features together, including but not limited to behavior, which are context and network-related. The framework provides the computation of a dynamic anomaly threshold for each user as per their profile age, i.e., profile maturity. It also has a real-time scoring and risk assessment method designed for large enterprise authentication systems. The efficacy of the framework is verified through evaluation on a large-scale enterprise authentication dataset comprising approximately 72 million records from over 2 million users, as described in detail in Section 3.4.1. The proposed framework contributes to an effective and secure authentication system using dynamic and hybrid anomaly detection.

The key novelty of this work lies in the joint integration of (i) per-user adaptive thresholds, (ii) profile-age-aware anomaly weighting, and (iii) a hybrid multi-model aggregation strategy designed for real-time authentication environments.

Section 3 explains a detailed description of the proposed Hybrid Anomaly Detection framework, including architectural design, anomaly detection methodologies, computation of the Hybrid Anomaly Score and threat model. Section 4 reports the experimental results and performance evaluation of the proposed approach, which, in-turn, is supported by hypothesis-driven ablation studies and comparative analysis with existing approaches. In the end, Sections 5 and 6 summarize the key findings and outline potential directions for future research, respectively.

2. Related Works

A modern authentication system requires more advanced methodologies of cyber-security. Researchers have developed methods of anomaly detection based on different factors like geography, devices, patterns of user activity, etc. [8,12–15]. It has been proven that anomaly detection is an important component in the Risk-Based Authentication (RBA) mechanism that helps to find a proper balance between safety and comfort [16]. This section reviews the available approaches and studies that explore the combination of these methodologies and highlights key challenges with respect to their practical use.

Researchers [8,17] studied anomaly detection based on behavioral analytics to improve user authentication mechanisms. However, the approach lacks generalizations that can constrain practical implementation. The models used in these papers learn too much from small and uniform datasets, making them unable to adapt well to new, diverse or vast authentication data, thus demonstrating reduced robustness. The paper [8] makes use of deep learning models for anomaly detection. Although powerful, however, these models are very resource-heavy and also difficult to integrate with other systems, resulting in limitations of their use in large or real-time scenarios. This makes it challenging to build models that are accurate and scalable for deployment over large-scale enterprise applications.

Iskhakov et al. (2023) [14] introduced an ML-based model designed to identify abnormal user behaviors in cyber-physical environments. The algorithm is based on common

audit logs and browser fingerprints to identify unusual user behavior. Although the function works well on smaller datasets, the model faces challenges with limited feature diversity and scalability issues when deployed in large real-time environments.

Budzys et al. (2024) [18] proposed deep learning-based models to identify threats in the authentication system of critical infrastructure. Similarly, Prem Sai (2024) [12] proposed an RBA protocol that adapts to real-time behavior and generates trust scores, combining dynamic evaluation and anomaly detection to fill gaps such as poor threat detection and excessive false positives in traditional systems.

Advanced approaches involve the use of unsupervised and semi-supervised machine learning, clustering, deep learning autoencoders, and statistical outlier detection for per-user anomaly detection [5,8,19]. The introduced methodologies, however, improve detection accuracy, but face challenges in computation requirements, data labeling, and effective aggregation of anomaly attributes.

Several studies leveraging both behavioral and contextual factors have also contributed to the development of adaptive authentication systems to enhance security. For instance, Shaik Vaseem Akram et al. (2022) [20] proposed a framework that collected user behavior through the integration of mouse and keystroke dynamics along with contextual information such as browser, device, and location data to build a comprehensive user profile. Several studies mentioned in [2,12,21] have also emphasized the importance of adaptive trust scores and dynamic risk weighting to continuously evaluate authentication trustworthiness within Identity and Access Management (IAM) systems.

Recent studies focused on using machine learning approaches and ensembles of machine learning algorithms in order to solve the problem of anomaly detection in authentication. Methods like autoencoders and recurrent neural networks helped to learn more about users' behavioral representations and spot subtle deviations [15,22,23]. Ensemble techniques like Isolation forest helped to develop more robust models for anomaly detection [24,25]. Despite those advancements, some problems still remain unresolved—the problem of model interpretability, increased computations, inability to adjust to the change of user's behavioral baseline in a particular case. Nevertheless, current approaches of behavior-aware-based authentication systems incorporate different contextual and behavioral factors related to users. However, there are some limitations in those approaches.

The following subsections present a comparison with existing adaptive RBA systems, highlight critical gaps, and describe the motivation and design rationale of the proposed framework.

2.1. Comparison with Recent Adaptive RBA Systems

Recent adaptive RBA systems make use of various types of behavioral and contextual information in order to analyze and evaluate the risk level for a particular authentication attempt. Usually, they use some global or semi-adaptive risk scoring function that calculates the risk level as a combination of anomaly detectors' scores based on such aggregated information as devices, geolocation, and behavior patterns [2,3,6]. Though they significantly increase safety in comparison to standard MFA, they do not allow personalization and adaptation to the change in user behavior over time.

The proposed solution is distinguished from existing systems in a number of ways: first, it makes use of per-user deviation thresholds ($TD_{u,i}$) and allows detecting anomalies for each particular user independently. Second, it is aware of the maturity of user profiles, i.e., adapts automatically to different behaviors of users based on the period of time when the user was registered in the system. Third, it relies not on a single anomaly detector but on an ensemble of them with an analytical and weighted criticality factor.

All design decisions have been made in order to overcome existing problems in RBA solutions. In addition, it helps to provide better support for new users and avoid unnecessary false alerts for established ones. Additionally, the proposed model is scalable and robust in large-scale environments. Moreover, while other approaches try to improve models' precision by making them complicated, the proposed solution tries to achieve the balance between performance and precision in order to achieve a fast response time (about milliseconds).

2.2. Critical Gaps in Existing Approaches

The efficacy of the user-level anomaly detection framework for user authentication systems relies heavily on historical behavioral data associated with individual users. The users are categorized as NEW, BUILDING and ESTABLISHED based on their profile age in the authentication system. The different profile ages actually define the profile maturity of the users under an authentication system. Users under the NEW category will have limited behavior history data, which may cause high anomaly scores and more false positives. Users under the BUILDING category will have moderate behavior data, while Users under the ESTABLISHED category will have sufficient behavior data. Most of the anomaly detection frameworks don't clearly consider these categories, leading to inconsistent results.

The critical problem with existing frameworks is that they use "one-size-fits-all" rules for all users, no matter how mature their profiles are. This often causes issues for NEW users, while small changes in behavior from ESTABLISHED users might not. Furthermore, the deviation computation of several systems does not rely on per-user behavioral baselines but relies on population-level statistics that obscure individualized behavior.

From a deployment perspective, existing anomaly detection frameworks face additional challenges, including elevated false-positive and false-negative rates in dynamic behavioral environments [8], privacy considerations related to behavioral and biometric data handling [5,26], scalability constraints under real-time per-user modeling requirements, and user experience degradation caused by excessive authentication prompts or latency [7,12,27,28]. These limitations collectively motivate the need for a more adaptive, profile-aware, and scalable anomaly detection framework.

Table 1 summarizes the fundamental limitations of existing methods and how the proposed framework addresses them.

Table 1. Comparison of existing approaches and the proposed framework.

Limitation	Existing Approaches	Proposed Framework
Static weights	Uniform weighting across users	Per-user profile maturity based weighting (α_u)
Lack of integration	Single-dimension anomaly detectors	Unified six anomaly detection aggregation
Limited scalability	Accuracy decline at scale	Stable performance on 72M records
Manual thresholds	Expert tuned static parameters	Dynamic thresholds
Limited real-time capability	60–139 ms latency	2–3 ms inference latency

2.3. Algorithm Motivation and Design Rationale

To address the identified limitations of the existing "one-size-fits-all" framework and elevated false positive and false negative rates in dynamic behavioral environments, the multi-model hybrid anomaly detection framework is proposed, which includes the calculation of the Hybrid Anomaly Score (HAS) with respect to wide categories of anomaly detection approaches as explained in Section 3. The framework illustrates the calculation of profile deviation score generated by different anomaly detection methods, and derives an algorithm to compute the consolidated HAS considering their impact on user authentication scenarios. The HAS is then used to assign a risk level through a three-tier

labeling mechanism: Green (normal behavior), Yellow (irregular but low-risk activity), and Red (high-risk anomaly). This ensemble approach focuses on the calculation per-user and system-wide anomaly scoring.

We build the architecture of our framework with three adaptive layers for computing the consolidated HAS: (1) computation of dynamic per-user profile deviation score, respectively, to the type of anomaly, (2) computing system-wide anomaly rates with respect to the user profile age; and (3) identifying criticality multipliers subject to the anomaly type. Our architecture effectively addresses the critical research gaps highlighted in [5], i.e., improving methods to aggregate and analyze multi-model behavior with per-user risk scoring.

The selection of multiple anomaly detection methodologies in the proposed framework is motivated by their complementary strengths reported in prior studies. Different machine learning models and ensembles can be employed to achieve good results. Statistical methods can be used to find stable distributions, such as clustering methods—non-linear and multi-dimensional deviations, tree-based—and multi-dimensional patterns [5,24]. Neural networks were successful in finding such attacks, such as DDoS or login attempts from multiple locations simultaneously [22]. Using those methods in combination, we will be able to achieve the desired results.

3. Hybrid Anomaly Detection Framework

This section presents the overall architecture of the proposed Hybrid Anomaly Score (HAS) framework, including the threat model, anomaly detection methodologies, and the formulation of personalized thresholds. The framework is designed to integrate multiple behavioral signals while ensuring scalability and real-time performance.

3.1. Threat Model

The threat model for the proposed framework assumes that the authentication system operates under a ZTA architecture. It means that all authentication processes should take place on constant evaluation of the user's identity, contextual and behavioral information in order to detect anomalies that would indicate potential unauthorized access to resources. At the same time, minimizing false positive detections while maintaining real-time operability.

3.1.1. Adversary Model

Adversary in this threat model is an authenticated, possibly malevolent actor who performs authentication processes within boundaries of the system. This may mean either an external attacker using legitimate credentials or an insider performing some actions inappropriately.

Thus, the adversary's objective is to perform the authentication process and not be caught by the anomaly detection algorithm. This directly challenges the detection capability of static and single-model systems.

It is assumed that the adversary possesses the following capabilities:

- Use of valid or partially compromised credentials;
- Authentication attempts from unfamiliar devices, locations, or networks;
- Low-rate or distributed login attempts to evade rate-limiting mechanisms;
- Partial imitation of legitimate user behavior patterns.

Multiple anomaly detection methodologies are used to explicitly model these behaviors. Such as, behavioral imitation is addressed through time-based and device fingerprinting models, while automated attacks are captured through login failure and IP anomaly detection.

It is assumed that the adversary does not have the capability to:

- Modify server-side authentication logs;

- Compromise the anomaly detection infrastructure;
- Perform full system-level breaches beyond credential misuse.

This adversary model ensures that the proposed Hybrid Anomaly Score (HAS) framework is designed to detect realistic and practically relevant authentication threats.

3.1.2. Threat Scenarios

The proposed framework addresses the following indicative threat scenarios:

Credential misuse: Access attempts that deviate from historical user behavior.

Lateral access attempts: Unauthorized access to applications or services using valid credentials.

Insider misuse: Behavioral deviations of legitimate users beyond baseline.

Stealthy automation: Automated authentication that exhibits subtle deviations.

3.1.3. Threat-to-Detection Mapping

For better representation of real-life adversary's actions, each type of attack must be mapped to some anomaly detection algorithms. Such mapping is presented below:

- Credential Misuse: Detectable through anomalies in time-based behavior, login patterns and device fingerprint.
- Lateral Access Attempts: Detectable by location-based anomalies and by irregularities in login-types.
- Insider Misuse: Identified based on deviations in the behavioral and temporal models that capture changes in existing user patterns.
- Stealthy Automation: Identified based on anomalous login failures, compromised IPs, and temporal clusters that identify slow rate/distributed attacks.

The above mappings assure that the various elements of HAS contribute directly towards identifying different forms of malicious activity as per the threat model definition.

3.1.4. Scope Limitations

The following are out of scope for this research effort: software weaknesses in authentication modules, physical endpoint tampering, denial-of-service attacks, and advanced persistent threats that leverage prolonged identity system compromise. These threats would require other complementary security measures along with anomaly-based authentication systems.

The above scenarios of adversarial activities inform the design of our Hybrid Anomaly Detection method. The various forms of attacks can be directly translated into anomalies and their corresponding feature space. For instance, credential misuse and insider threats are captured through behavioral deviation and login pattern analysis, while automated and distributed attacks are detected using IP reputation, login failure patterns, and temporal anomalies.

Furthermore, the dataset that we are using in our analysis involves several threat scenarios such as unauthorized login attempts, device switches, geographic divergence and abnormal user accesses. Therefore, the proposed Hybrid Anomaly Score (HAS) can be said to be aligned with practical adversary attacks.

3.2. Mathematical Notation and Assumptions

To make the discussion easy to understand and follow, all variables and assumptions that have been made in the Hybrid Anomaly Score (HAS) are clearly defined below.

Suppose $u \in \mathcal{U}$ is a user while $i \in \{1, 2, \dots, M\}$ is an index for different anomaly detection methodologies, where M is the total number of methodologies.

$PD_{u,i}$ is the profile deviation score for user u for methodology i , while $TD_{u,i}$ is the personalized deviation threshold of that particular methodology for the same user. In this way, a binary anomaly score $A_{u,i} \in \{0, 1\}$ is defined where $A_{u,i} = 1$ when $PD_{u,i} > TD_{u,i}$ otherwise 0. Furthermore, $\Delta_{u,i}$ is the normalized value of deviation severity.

In terms of statistical parameters, $\mu_{u,i}^{(H)}$ refers to the historical mean, and $\sigma_{u,i}^{(H)}$ refers to the standard deviation of the deviation score of the user for methodology i computed over a historical data window of size H . λ refers to a factor controlling deviation threshold sensitivity while $\epsilon > 0$ is just a positive infinitesimal.

In terms of aggregation, AR_{i,pa_u} refers to the anomaly rate of a given anomaly detection methodology i for a specific user cohort of profile age pa_u . The weight of a given methodology BW_i is denoted as a baseline weight, while $\hat{B}W_i$ represents normalized baseline weights. Criticality associated with methodology i is B_i while criticality multipliers $CM_{u,i}$ are calculated per user and per method.

The final output of the framework is the Hybrid Anomaly Score $HAS_u \in [0, 1]$, representing the overall anomaly risk associated with user u .

The formulation relies on the following assumptions. First, historical deviation scores are assumed to follow a stationary distribution within the observation window H . Second, the anomaly detection methodologies are treated as independent, allowing their contributions to be aggregated without interaction effects. Third, anomaly rates AR_{i,pa_u} are assumed to remain stable within each profile-age cohort. Finally, the inclusion of ϵ ensures numerical robustness in all computations.

3.3. Framework Overview

The proposed framework is designed to detect behavioral deviations and authentication abuse attempts defined under the assumed threat model, without requiring prior knowledge of specific attack signatures.

The approach implements a hybrid framework that computes the Hybrid Anomaly Score (HAS_u) for each authentication activity of the users. For the computation of the Hybrid Anomaly Score, the framework introduces the fundamentals of the dynamically computed profile deviation score threshold ($TD_{u,i}$), which is calculated per-user, and the criticality multiplier for every anomaly detection methodology (i). The value of the profile deviation score threshold ($TD_{u,i}$) is used to identify the anomaly of users' authentication activity per anomaly detection methodology (i). The framework introduces multiple types of anomaly detection methodologies, time-based user behavior anomaly, login failure anomaly, user login type anomaly, location-based behavior anomaly, compromised IP detection anomaly and device and mouse behavior anomaly. Each anomaly detection methodology is briefly described below.

Time-based behavior anomaly: This methodology identifies abnormal login activities of the user based on the activity time window. This monitors the user on when and how frequently a user accesses the account, flagging sudden deviations, such as late-night access or repeated attempts in a short period of time, thus indicating unusual or potentially unsafe activity.

Login Failure-Type Anomaly: This methodology aims at identifying suspicious login failures exceed their normal pattern, such as repeated invalid OTP, password attempts or unusual login failure attempt. A sudden spike beyond the expected limit is treated as possible malicious behavior.

User Login-Type Anomaly: This methodology detects unusual spikes in a user's number of login attempts whether successful or unsuccessful, over a certain time period, such as 15 min.

Location-Based behavior Anomaly: This methodology focuses on detecting unexpected changes in a user's login location, flagging logins from entirely different locations, especially when combination with repeated attempts or failures—as potential signs of unauthorized access or account compromise.

Compromised IP Anomaly: This method identifies the potential vulnerable IP addresses by inspecting several failed login attempts from the same IP within a short time window, indicating threats such as brute-force activity, credential stuffing, or bot-generated login attempts. These attack patterns correspond to the automated adversary behaviors defined in the threat model.

Device Fingerprinting and Mouse behavior Anomaly: This methodology aims to integrate user fingerprinting with user mouse behavior analysis along with device fingerprint to check whether a device is being used by actual user or by someone else. When a user who normally uses a certain device all the time starts to use a new device and performs weird mouse operations, then this is in line with partially compromised accounts and impersonated adversaries, as mentioned in the threat model.

3.3.1. Model Selection and Theoretical Complementarity

As far as anomaly detection methodologies are concerned, there are two main criteria for selecting appropriate methods, namely behavioral orthogonality and statistical complementarity. The idea is to find different anomaly detectors that work with different statistical assumptions to detect anomalies from various perspectives.

For time-based anomalies, clustering techniques (k-means), along with classifiers (SVM), are used because they are capable of finding periodic patterns in user data. As far as login failures are concerned, anomalies are detected using $\mu + 3\sigma$ deviation models since it works efficiently in detecting count-based anomalies having stable data distribution.

Isolation Forest is selected for modeling anomalies regarding login types and locations due to the high dimensional and sparse nature of the data. The geographical deviations are further enhanced using the haversine distance since it enables physical meaning in geographical deviations.

Detection of IP anomalies is achieved through neural networks and profile based blending which is capable of handling non-linear attacks like distributed or low rate credential stuffing. Detection of Device Fingerprint and Mouse Behaviors requires a combination of Random Forest, BIRCH clustering, MinHashLSH models to achieve anomaly detection of these two categories.

From a theoretical standpoint, the complementarity arises because these detectors work on different and orthogonal statistical properties. Statistical models capture distributional deviations, clustering methods capture structural patterns, tree-based models capture non-linear feature interactions, and neural networks capture complex latent representations. The combination of these models minimizes the correlation among failure modes of detectors and helps in enhancing model robustness against adversarial evasion attacks.

Thus, the proposed combination of these detectors is in line with previous research conducted in the field of ensemble anomaly detection [5,8,15]. By integrating multiple weakly correlated anomaly signals, the proposed Hybrid Anomaly Score (HAS) achieves improved reliability compared to any individual detector operating in isolation.

3.3.2. Framework Execution Pipeline

The proposed framework is divided into four phases: (i) training phase (multi-dimensional anomaly detection), (ii) real-time phase (anomaly detection per detection methodology), (iii) hybrid anomaly detection phase (computation of Hybrid Anomaly

Score) and (iv) risk categorization (decision layer). These phases operate in a strict pipeline order.

First, the training phase is executed offline using historical authentication data to estimate per-user deviation thresholds ($TD_{u,i}$) and profile-age-conditioned anomaly rates (AR_{i,pa_u}). This phase is performed periodically and does not participate in real-time decision making.

Next, during system operation, incoming authentication events are processed in the Real-Time Detection Phase, where methodology-specific features are extracted and profile deviation scores ($PD_{u,i}$) are computed for each anomaly detection model. These scores are immediately evaluated against the precomputed thresholds ($TD_{u,i}$) to generate anomaly indicators ($A_{u,i}$) and deviation severities ($\Delta_{u,i}$).

After computing the anomaly score in real-time phase, the next step is to perform the hybrid aggregation. This process requires normalization of weights along with the application of criticality multipliers to finally calculate the Hybrid Anomaly Score (HAS_u) for each user. This score is then used for risk categorization and downstream authentication decisions.

In conclusion, the proposed framework can be seen to operate as a three-step unidirectional pipeline as follows:

Training (offline) → Real-Time Detection → Hybrid Aggregation

This clears the separation between learning and inference which makes real-time inference possible and efficient through adaptive and data driven behavior.

Figure 1 provides a detailed representation of all the phases of the proposed framework.

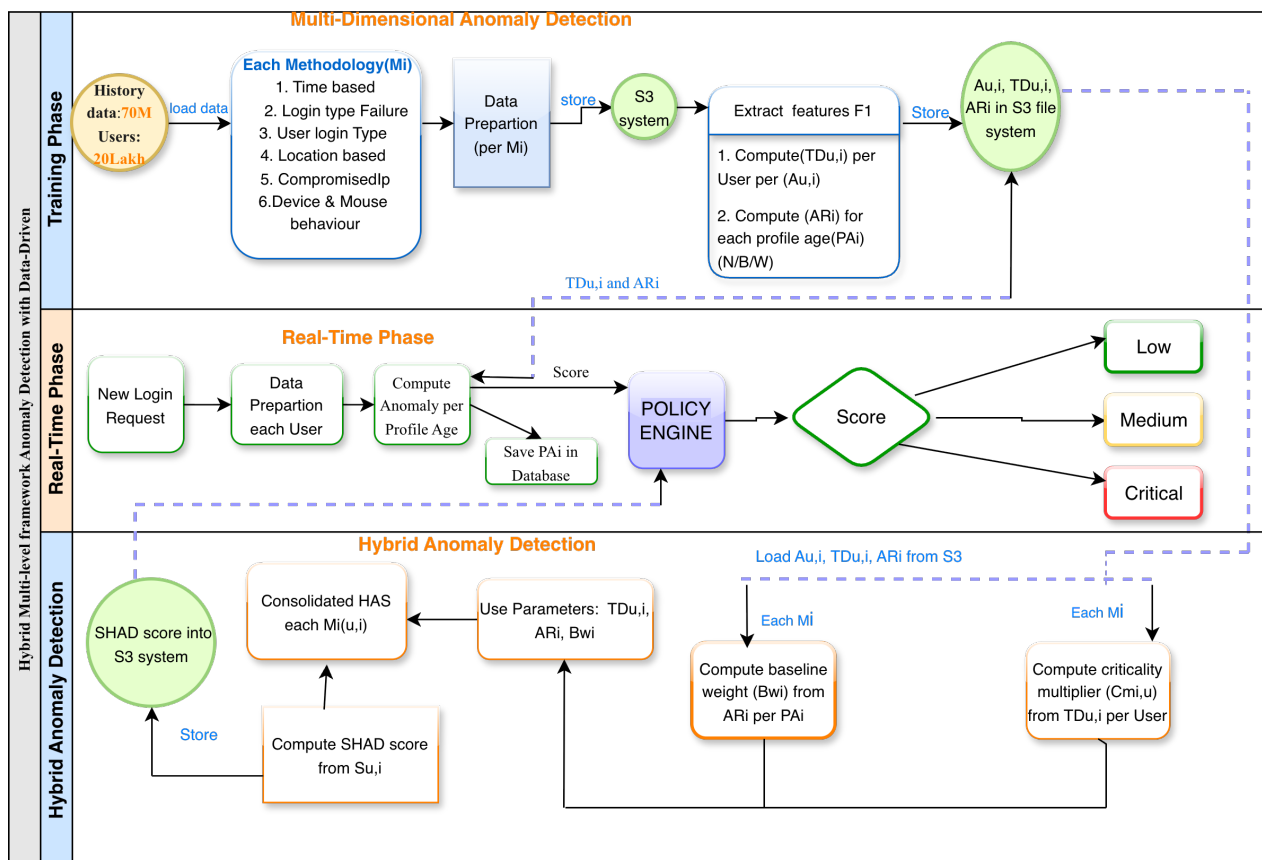


Figure 1. Overview of the proposed hybrid anomaly detection framework. Incoming authentication events are processed in real time to compute method-specific profile deviation scores ($PD_{u,i}$), which are evaluated against personalized thresholds ($TD_{u,i}$) to derive deviation severities ($\Delta_{u,i}$) and binary

anomaly indicators ($A_{u,i}$). Profile-age—conditioned anomaly rates are used to compute baseline weights ($B\hat{W}_i$), which, together with severity-scaled criticality multipliers ($CM_{u,i}$), are aggregated to produce a bounded Hybrid Anomaly Score (HAS) for user-level risk categorization. Solid arrows denote data flow, while dashed lines represent parameter sharing and dependency relationships across components.

3.4. Training Phase: Multi-Dimensional Anomaly Detection

This section describes the training phase of the proposed framework, including dataset characteristics, preprocessing steps, and model training procedures. The goal is to establish reliable user-specific behavioral baselines for subsequent real-time anomaly detection.

3.4.1. DataSet

A dataset used in this study is collected from large-scale enterprise application comprising approximately 72 million authentication records collected from over 2 million users over a six-month period. This dataset is used for training the models used in the study. Similar large-scale authentication datasets have been used in prior studies [6,9].

Before data analysis, anonymization of personally identifiable information (PII) was performed by hashing and tokenization to comply with data governance and data privacy policies. Due to confidentiality constraints, the dataset is not publicly available.

To improve reproducibility and transparency, we summarize the statistical properties of the dataset. Every authentication event carry behavioral, contextual, and system-level attributes extracted from enterprise authentication system logs. The dataset includes all types of user authentication-related activities, including login, logout, token-based authentication, password and password-less authentication, and geo-fencing events.

The dataset naturally captures the defined threat scenarios through authentication logs containing failed login attempts, device changes, geographic access variations, and abnormal login patterns. These signals correspond to the actual adversarial behaviors outlined in the threat model and provide the basis for labeling the anomalous events. The labels of the anomalies will therefore come as an output of a hybrid process involving the combination of rules, reported incidents/verification and statistical deviation analysis. This ensures that labeled anomalies reflect realistic authentication risks observed in production environments rather than synthetic or artificially injected anomalies.

3.4.2. Dataset Characteristics

The dataset is based on 36 feature dimensions and labeled using a hybrid methodology consisting of rule-based signals, verified security incidents along with statistical deviation from historical per user behaviors. The resulting dataset has highly skewed classes with a very small percentage of anomalies being identified through HAS based labeling as indicated in Table 2.

Table 2. Dataset Summary Statistics.

Metric	Value
Users	2,200,000
Raw Events	72,000,000
Feature Dimensions	36
Label Source	Hybrid rule-based and statistical approach
Anomalous Events (%)	3.2% (Derived from HAS-based labeling)
Normal Events (%)	96.8% (Complement of anomaly rate)
Missing Data (%)	1.4% (Calculated across all features)

The dataset captures multiple types of authentication anomalies aligned with real-world threat scenarios. These include repeated login failures indicative of brute-force attacks, abnormal login frequency patterns associated with automated attacks, unusual geographic login locations suggesting possible credential misuse, and device fingerprint inconsistencies indicating potential session hijacking. These anomaly patterns are consistent with the defined threat model and ensure that the evaluation reflects realistic authentication risks encountered in enterprise environments.

Using this dataset, the next section discusses the experimental setup for evaluating the HAS model.

3.4.3. Model Training

There are three phases involved in model training which include data preparation, estimation of personalized threshold of the computed profile deviation score ($TD_{u,i}$) and calculation of anomaly rates per anomaly detection method (AR_i) on the basis of per-user profile age. Model training is performed for each anomaly detection methodology M_i independently.

Data Preparation: This phase involves partitioning of authentication records into daily batch groups. This data is processed to get the required feature set for specific anomaly detection method. The batch data represents historical behavior data used in estimating deviation characteristics for individual users.

Data Cleaning and Preprocessing Pipeline: In order to improve the accuracy and reliability of the detection algorithms, a pre-processing pipeline was developed for processing the authentication logs before any features were extracted from them. This included the removal of noise from the data, which included duplicates, partial entries, and corrupt entries. Authentications with invalid sessions or inconsistent state transition patterns (e.g., logins without prior initiation) were disregarded.

Missing data was addressed using a mixed imputation approach tailored to each type of feature. For quantitative features (e.g., login frequencies and rates of failures), imputation was conducted using statistical calculations on a rolling window basis (such as median and mean) per user. Meanwhile, qualitative variables (e.g., device types, browsers, and location), were given an “unknown” category. Features with too many missing values were removed from the analysis. Outliers were treated with extreme caution, without discarding them to maintain the integrity of potentially malicious behavior. Extreme outliers exceeding the 99th percentile were either winsorized or normalized with statistical approaches (such as robust scaling).

Lastly, features were standardized and converted into models’ input formats according to their requirements. This entire process guarantees that the final data set reflects realistic user behavior in terms of anomalies despite potential inconsistencies.

Computation of User-Based Profile Deviation Score: Calculating $PD_{u,i}$ requires a deviation from user-based historical behavioral baselines, expressed by the equation below. The calculation is done for each methodology M_i and per user basis. The deviation threshold $TD_{u,i}$ is then derived from the empirical distribution of historical deviation scores for corresponding user u . This enables personalization without reliance on global thresholds. This is followed by the exclusive offline validation of detection behavior using precision and recall metrics which is not used for direct threshold tuning, thereby reducing the risk of overfitting.

Dynamic Threshold Estimation: For each user u and methodology M_i , a personalized deviation threshold $TD_{u,i}$ is computed from historical deviation scores using a rolling observation window:

$$T_{u,i} = \mu_{u,i}^{(H)} + \lambda \cdot \sigma_{u,i}^{(H)} \quad (1)$$

where $\mu_{u,i}^{(H)}$ and $\sigma_{u,i}^{(H)}$ denote the mean and standard deviation of historical deviation scores over window H , and λ controls detection sensitivity.

The sensitivity parameter λ controls the strictness of anomaly detection. In this study, $\lambda = 3$ is used, corresponding to the standard 3σ rule. This value balances detection sensitivity and false positive control and was validated through offline experiments.

Anomaly Rate Computation: The effectiveness of the anomaly detection algorithm depends upon accurately estimating the anomaly rate of the system with respect to the profile age of the users. To validate the anomaly and optimize the anomaly detection, each user's behavioral deviation is expressed as a percentage-based anomaly rate, representing the proportion of users falling within the anomaly. A higher degree of deviation in recorded user behavior may correspond to a non-high anomaly rate. The anomaly rate is performed with respect to the NEW, BUILDING and ESTABLISHED user profile age categories.

Table 3 refers to the feature set and models/algorithms used for the computation of profile deviation score threshold and anomaly rate for each anomaly detection methodology.

Table 3. Summary of the six anomaly detection methodologies used in the proposed hybrid framework. Each methodology M_i operates on a distinct feature space and underlying detection model to compute a continuous profile deviation score $PD_{u,i}$ for incoming authentication events during the real-time phase. These deviation scores are subsequently evaluated against personalized thresholds $TD_{u,i}$ and aggregated into the Hybrid Anomaly Score (HAS).

Methodology	Key Features Used	Model/Algorithm
Time-based Behavior	Time-of-day, weekday, login frequency, user, date	KMeans + SVM
Login Failure Type	User, date, and multiple failure counts including OTP failures, password failures, username attempts, geo-fencing failures, and resend attempts	Statistical Deviation Modeling ($\mu + 3\sigma$ Rule): A distance of three standard deviations σ from the mean μ .
User Login Type	Failure counts (e.g., MF failure, credential failures), and success counts	Isolation Forest
Location-Based User Behavior	Location states, timestamps, travel distance and speed, historical patterns	Haversine + Isolation Forest
Compromised IP	Failure count, users per window, historical IP usage, IP failure profile	Neural Network + Profile Blending
Device Fingerprinting & Mouse Behavior	Device ID, OS/CPU/GPU, screen size, browser type, geo-IP, clicks, scrolls, and mouse movement patterns	Random Forest + BIRCH + MinHashLSH

3.5. Real-Time Phase

The real-time phase of the proposed framework is responsible for ingesting live authentication events, performing feature extraction, computing per-methodology deviation scores, and generating anomaly indicators for each authentication activity. This phase operates independently of offline model training and does not require ground-truth anomaly labels.

Dataset: Authentication events are streamed continuously from production systems using a scalable event-ingestion pipeline and processed under a fixed 15-min sliding window configuration, reflecting operational constraints in enterprise authentication environments. Each 15-min window contains approximately 3–5 K authentication events with authentication activities of 20 K users.

Data Preparation: For each incoming authentication event, methodology-specific feature extraction is performed according to the six anomaly detection approaches summarized in Table 3. In parallel, the user profile-age category ($pa_u \in \{\text{NEW, BUILDING,}$

ESTABLISHED}) is identified using historical metadata. The extracted features are then forwarded to the corresponding anomaly detection models for real-time deviation analysis.

Computation of Profile Deviation Score For each incoming authentication event associated with user u , a continuous profile deviation score $PD_{u,i}$ is computed independently for each anomaly detection methodology M_i . The deviation score quantifies the degree to which the observed behavior deviates from the user's historical baseline under the corresponding feature space.

The deviation score is derived from the output of the underlying detection model, such as distance from learned temporal clusters, statistical deviation from historical failure distributions, isolation forest decision scores, geographic travel anomalies, neural network-based IP risk estimation, or device and behavioral fingerprint mismatches.

Deviation Severity Computation

To quantify the magnitude of anomalous deviation beyond the personalized threshold, a normalized deviation severity is computed as:

$$\Delta_{u,i} = \max\left(0, \frac{PD_{u,i} - TD_{u,i}}{TD_{u,i} + \epsilon}\right) \quad (2)$$

The severity term enables dynamic scaling of anomaly impact while preserving robustness against minor fluctuations in user behavior.

Binary Anomaly Decision

The binary anomaly indicator for user u under methodology M_i is defined as:

$$A_{u,i} = \begin{cases} 1, & \text{if } PD_{u,i} > TD_{u,i} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

where $A_{u,i} = 1$ denotes anomalous behavior and $A_{u,i} = 0$ denotes normal behavior.

3.6. Computation of Hybrid Anomaly Score (HAS)

This section presents the formulation of the Hybrid Anomaly Score (HAS_u), which aggregates heterogeneous anomaly signals to derive a unified user-level risk score. The computation integrates information obtained during both the model training and real-time detection phases.

The notation used in the HAS formulation follows Section 3.2. Only key relationships are summarized below for continuity.

$TD_{u,i}$: Dynamically learned per-user deviation threshold for anomaly detection methodology M_i .

AR_{i,pa_u} : Anomaly rate associated with methodology M_i within the user profile-age category

$pa_u \in \{NEW, BUILDING, ESTABLISHED\}$.

$A_{u,i} \in \{0, 1\}$: Binary anomaly indicator for user u under methodology M_i .

BW_i : Baseline weight assigned to methodology M_i , derived from anomaly rate statistics.

$CM_{u,i}$: Criticality multiplier reflecting the relative security impact of anomalies detected by M_i .

$\Delta_{u,i}$: Deviation severity for methodology M_i with respect to user u . All deviation severities used in the HAS_u computation are obtained from the real-time deviation analysis described in Section 3.5.

Algorithm

The HAS formulation integrates outputs from the real-time phase (i.e., $PD_{u,i}$, $A_{u,i}$, $\Delta_{u,i}$) and statistical estimates obtained during training (i.e., $TD_{u,i}$ and AR_{i,pa_u}). This sep-

ation ensures that the scoring mechanism remains deployable in real-time without requiring re-training.

Step 1: Baseline Weight Computation

The baseline weight for each anomaly detection methodology is derived from the observed anomaly rate for the corresponding user profile-age cohort:

$$BW_i = \frac{1}{AR_{i,pa_u} + \epsilon} \quad (4)$$

The weights are normalized across all M methodologies to ensure comparability:

$$\hat{BW}_i = \frac{BW_i}{\sum_{j=1}^M BW_j} \quad (5)$$

Step 2: Criticality Multiplier Computation

Each anomaly detection methodology is associated with a baseline business-criticality constant B_i , reflecting the relative security impact of anomalies detected by that method. The criticality multiplier is dynamically scaled on deviation severity:

$$CM_{u,i} = B_i \cdot (1 + \Delta_{u,i}) \quad (6)$$

where $\Delta_{u,i}$ denotes the normalized deviation severity of the detected deviation, where $B_i \in \{1.0, 1.2, 1.5, 2.0\}$, representing four risk tiers of “low”, “medium”, “high” and “critical”.

The business criticality constant B_i was calculated based on the security impact of the anomalies. Time-based behavior and user login type anomalies were labeled as “medium” risk ($B_i = 1.2$). Login failure, location-based and device fingerprinting were classified as “high” risk ($B_i = 1.5$). Compromised IP detection was regarded as “critical” ($B_i = 2.0$), as it has the most direct relation to credential abuse. Meanwhile, the lowest criticality B_i value of 1.0 applies to low-risk alerts.

Step 3: Method-Level Contribution

Each anomaly detection methodology contributes to the hybrid score calculation as follows:

$$S_{u,i} = A_{u,i} \cdot \hat{BW}_i \cdot CM_{u,i} \quad (7)$$

Step 3(a): Normalization of Method Contributions In order to cap the contribution of individual methods and reduce the possibility of having one single anomaly detector dominate others, the method-level score is constrained as follows:

$$S_{u,i}^{\text{norm}} = \min(1, S_{u,i}) \quad (8)$$

Step 4: Final Hybrid Anomaly Score

The final Hybrid Anomaly Score for user u was calculated by summing up the contributions of all the anomaly detection algorithms:

$$HAS_u = \sum_{i=1}^M S_{u,i}^{\text{norm}} \quad (9)$$

The aggregated score is bounded to ensure $HAS_u \in [0, 1]$:

$$HAS_u = \min(1, HAS_u) \quad (10)$$

This ensures that the HAS remains as a probabilistic risk score. Conditional independence among all anomaly detection algorithms is assumed during aggregation. Although

there is mild dependence among algorithms, the clipping and normalization operation in Step 3 (a) prevents one algorithm from dominating others. The algorithm to compute the Hybrid Anomaly Score (HAS) is shown formally in Algorithm 1.

Algorithm 1 Dynamic Hybrid Anomaly Score (HAS) Computation

Require: Authentication event e_u for user u
Require: Historical deviation statistics $\{\mu_{u,i}^{(H)}, \sigma_{u,i}^{(H)}\}$
Require: Profile-age conditioned anomaly rates $\{AR_{i,pa_u}\}$
Require: Business criticality constants $\{B_i\}$
Ensure: Hybrid Anomaly Score $HAS_u \in [0, 1]$

```

1: for  $i = 1$  to  $M$  do
2:   Compute profile deviation score  $PD_{u,i}$ 
3:    $TD_{u,i} \leftarrow \mu_{u,i}^{(H)} + \lambda \sigma_{u,i}^{(H)}$ 
4:    $A_{u,i} \leftarrow \mathbf{1}(PD_{u,i} > TD_{u,i})$ 
5:    $\Delta_{u,i} \leftarrow \max\left(0, \frac{PD_{u,i} - TD_{u,i}}{TD_{u,i} + \epsilon}\right)$ 
6: end for
7:  $BW_i \leftarrow \frac{1}{AR_{i,pa_u} + \epsilon}$  for all  $i$ 
8:  $\hat{BW}_i \leftarrow \frac{BW_i}{\sum_{j=1}^M BW_j}$ 
9: for  $i = 1$  to  $M$  do
10:   $CM_{u,i} \leftarrow B_i(1 + \Delta_{u,i})$ 
11:   $S_{u,i}^{norm} \leftarrow \min(1, A_{u,i} \hat{BW}_i CM_{u,i})$ 
12: end for
13:  $HAS_u \leftarrow \min\left(1, \sum_{i=1}^M S_{u,i}^{norm}\right)$ 
    return  $HAS_u$ 

```

3.7. Risk Categorization

HAS_u refers to a continuous measure of a user's riskiness within the interval of $[0, 1]$. Based on the score calculated in the previous section, the step is to categorize the users as Low-risk, Medium-risk, and High-risk based on the score generated.

Specifically, this step converts a user's continuous risk score into a discrete risk rating, in which three tiers ("Low", "Medium" and "High") will be defined for different decisions. In addition to enabling decisions based on the risk score, the following approach allows for a dynamically computed risk categorization architecture that can seamlessly integrate with the RBA framework.

The risk categorization function is defined as:

$$\text{Risk}(u) = \begin{cases} \text{Low Risk}, & 0 \leq HAS_u \leq \tau_1 \\ \text{Medium Risk}, & \tau_1 < HAS_u \leq \tau_2 \\ \text{High Risk}, & HAS_u > \tau_2 \end{cases}$$

Specifically, two thresholds, τ_1 and τ_2 , are selected from the empirical distribution of HAS_u values.

The low-risk category consists of all authentication events that have only small deviations that still fit well with historical user behavior. These events likely do not need extra authentications. Events in the medium-risk category, however, indicate moderate deviations, thus requiring more verification. On the contrary, the high-risk category consists of events with large deviations, suggesting malicious activity.

Percentile-Based Risk Boundary Selection: As aforementioned, the values of the thresholds τ_1 and τ_2 are dynamically calculated instead of being pre-assigned. These threshold values are computed using percentile-based cutoffs based on the distribution of

historical enterprise authentication data. From the distribution chart above, it can be seen that τ_1 is the 80th percentile, while τ_2 is the 95th percentile of the distribution of HAS_u . This figure reveals a disproportionate distribution in enterprise authentication data, wherein the large number of HAS_u values are at low risks, whereas few HAS_u values correspond to high-risk activities.

This dynamically computed risk categorization architecture facilitates seamless integration of risk scoring into an RBA policy framework.

Interpretation: HAS can be thought of as an aggregated measure of weighted anomalies in which weights indicate the reliability of detection algorithms, and multiplicand indicates security criticality. Both characteristics make the algorithm statistically robust and interpretable.

Formalizing risk categorization as a decision layer completes the end-to-end anomaly detection and response workflow of the proposed framework.

4. Evaluation Results and Validation

This section reports a comprehensive practical evaluation of the proposed Hybrid Anomaly Score (HAS) framework. The evaluation of the framework is done using large-scale enterprise authentication data. The evaluation is designed to validate the research questions and hypotheses as outlined in Section 1.1. The evaluation is focused on (i) the effectiveness of personalization, (ii) stability across heterogeneous user profile age group, and (iii) robustness and generalizability of the aggregation strategy.

4.1. Experimental Setup

The proposed approach was implemented in Python (version 3.12) and was deployed on a distributed infrastructure with horizontal scaling capability, ensuring consistent millisecond-level inference latency under concurrent authentication workloads.

The experiments are conducted on 72 million authentication events collected over six months from an enterprise-level authentication system of 2 million users. For the evaluation of the temporal stability and scalability of the framework, cumulative training windows of 1, 2, 4, and 6 months are used, while performance is measured on a fixed test set.

To maintain data privacy, all authentication records are anonymized prior to analysis, and no personally identifiable information (PII) was stored or processed.

Method-wise data coverage: To make sure that each anomaly detection methodology is trained and evaluated on a sufficient and representative volume of data, Table 4 summarizes the number of authentication records processed per methodology across different training windows, along with the corresponding user coverage.

Table 4. Volume of authentication records used in HAS computation across different time windows.

Methodology	1 Month	2 Months	4 Months	6 Months	Users
Login Failure Type	2.35M	3.35M	5.35M	7.35M	1.2M
User Login Type	28.21M	35.21M	49.21M	63.21M	1.8M
Compromised IP	2.01M	2.58M	3.73M	4.88M	0.12M
Time-Based Behavior	30.15M	35.15M	45.15M	55.15M	1.9M
Location-Based Behavior	10.47M	13.11M	18.39M	23.67M	1.5M
Device Fingerprint	27.90M	35.15M	49.65M	64.15M	2.2M

4.2. Temporal Stability Across Training Windows

Table 5 reports the quantitative anomaly detection performance of the HAS framework under increasing training history data. As historical behavioral data increases, precision and recall improve consistently, while the false positive rate (FPR) decreases monotonically.

Figure 2 demonstrates the distribution of user-level HASs computed from the six-month training period. A uniform dispersion of scores indicates that the aggregation of anomalies did not create significant clusters or artificial threshold boundaries as the volume of data grows. The finding suggests that the proposed framework did not overfit short-term behavioral patterns, but maintains stable detection behaviors as data history expands.

Table 5. Month-wise performance of HAS with increasing training data volume.

Training Window	Records (M)	Users (M)	Precision	Recall	FPR
1 Month	101.09	0.42	0.91	0.87	0.06
2 Months	124.55	0.98	0.93	0.89	0.05
4 Months	171.48	1.62	0.95	0.91	0.04
6 Months	218.41	2.00	0.96	0.92	0.03

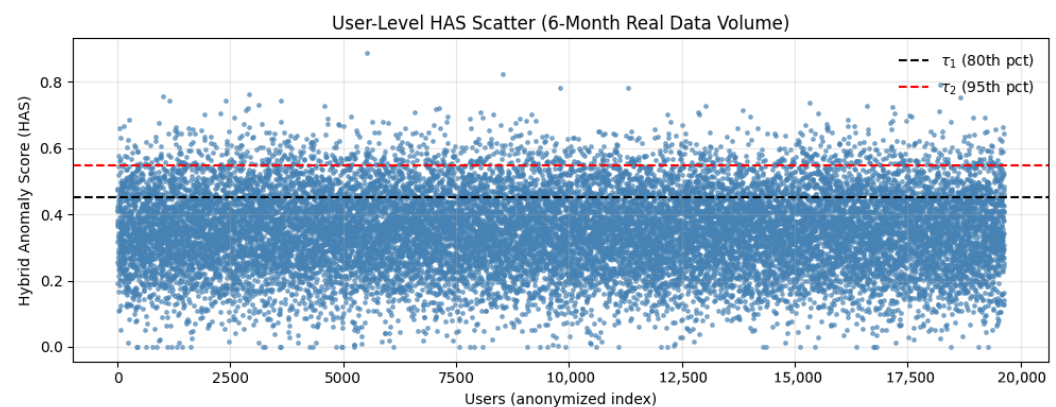


Figure 2. User-level Hybrid Anomaly Score (HAS) distribution over a six-month training window.

4.3. Profile-Age-Wise Performance Analysis

The performance of anomaly detection is analyzed across the user profile age i.e., NEW, BUILDING, and ESTABLISHED. As shown in Table 6, the HAS framework maintains high recall for NEW users while substantially reducing false positives for ESTABLISHED users, confirming effective personalization.

Table 6. HAS performance across user profile-age cohorts.

Profile Age Cohort	Users (K)	Records (M)	Precision	Recall	FPR
NEW (1–7 d)	410	9.4	0.90	0.91	0.07
BUILDING (8–30 d)	680	21.7	0.94	0.92	0.05
ESTABLISHED (>30 d)	910	40.9	0.97	0.93	0.02

The lower false positive rate observed for ESTABLISHED users is primarily due to the availability of stable and sufficiently rich behavioral history. With increasing maturity levels in user profiles, the accuracy of threshold estimation ($TD_{u,i}$) increases, and thus false positives are minimized.

However, NEW users have only a small amount of historical information window (1–7 days); hence, their deviations ($PD_{u,i}$) show higher variance and higher false positives. Thus, it shows that a profile age-aware approach helps the model discriminate between legitimate deviations and anomalies when user profiles mature. This phenomenon offers another layer of information on the efficiency of the framework developed, which further validates hypothesis H2.

On the other hand, when the profile maturity becomes lower, normal variations could contribute to deviations greater than the threshold $TD_{u,i}$. This effect is inevitable because

of the increased variance in statistical estimates. Hence, as profile maturity decreases, $PD_{u,i}$ might be greater than $TD_{u,i}$ resulting in increased false positive detections.

As the observation window expands for BUILDING and ESTABLISHED users, the estimations of mean and variance become stable and consistent. This behavior is consistent with the bias–variance trade-off in statistical estimation, where shorter histories introduce higher variance, while longer histories provide more reliable behavioral baselines. The above observation provides further validation of the hypothesis H2, that the maturity of users’ profiles is related to statistical stability in anomaly detection.

Given the large scale of the dataset used in this study (>70 million authentications), the observations were statistically significant, and additional significance testing is unnecessary.

4.4. Comparison with State-of-the-Art Deep Learning and Ensemble Baselines

In addition to benchmarking the Hybrid Anomaly Score (HAS) model against popular modern anomaly detection baselines, we conduct experiments to compare our model against two baseline approaches, namely deep learning and ensemble-based approaches. First, the Autoencoder-based anomaly detection model, which learns compressed representation of user behavior in authentication and performs anomaly detection by reconstruction loss. Second, the Isolation Forest ensemble model performed on unified feature space.

The Autoencoder anomaly detector is trained on normal authentication events and learns the minimum reconstruction loss, where any anomaly is detected when reconstruction loss exceeds a learned threshold. The Isolation Forest is a well-known baseline anomaly detector, which works by isolating anomalies by creating partitions.

Table 7 presents the comparative performance of baselines with the HAS framework.

Table 7. Comparison with deep learning and ensemble baselines.

Method	Precision	Recall	FPR	Latency (ms)
Autoencoder (DL)	0.91	0.88	0.06	18–25
Isolation Forest (Ensemble)	0.93	0.89	0.05	10–15
Proposed HAS Framework	0.96	0.92	0.03	2–3

The results demonstrate that while deep learning and ensemble methods provide competitive detection performance, they exhibit higher inference latency and limited adaptability to per-user behavioral baselines. In contrast, the proposed HAS framework achieves superior precision and lower false positive rates while maintaining millisecond-level latency. This highlights the effectiveness of hybrid aggregation with personalized thresholds over monolithic learning-based approaches in large-scale authentication environments.

4.5. Ablation Analysis of HAS Components

To isolate the contribution of individual design components within the proposed Hybrid Anomaly Score (HAS) framework, we conduct a series of controlled ablation experiments. In each experiment, a single component is removed or replaced while all other components remain unchanged, enabling causal analysis of its impact on overall system behavior.

We perform a systematic ablation study to evaluate how much each part of our proposed framework contributes to performance by either removing or modifying key elements of the proposed framework, and keeping everything else constant. The ablation configurations that we evaluate include: (i) Global thresholding instead of personalized thresholds, (ii) Removing profile-age-aware weighting, (iii) Removing criticality multipliers, (iv) Single-model detection (no hybrid aggregation), and (v) Full proposed framework (all components enabled).

For this study, the proposed framework modifies a single component on each test, with all other parts being held constant. The same dataset, training window, and evaluation protocol are used across all experiments to ensure controlled and fair comparison.

4.5.1. Effect of Personalized Thresholding (Ablation A1)

In this experiment, the personalized threshold $T_{u,i}$ is replaced with a global threshold computed using population-level statistics.

Personalized per-user deviation thresholds are compared against a static global threshold baseline. As shown in Table 8, static thresholding produces significantly higher false positive rates, particularly for users with limited behavioral history. Personalized thresholds adapt to individual baselines, substantially reducing false positives without loss of recall, validating Hypothesis H1.

Table 8. Impact of personalized thresholding versus static thresholds (Ablation A1).

Metric	Records (M)	Static	Personalized	Relative Change
Overall FPR	72.0	12.4%	5.9%	−52.4%
FPR (NEW Users)	9.4	18.7%	7.2%	−61.5%
Precision	72.0	0.86	0.96	+0.10
Recall	72.0	0.92	0.92	No degradation

4.5.2. Effect of Profile-Age-Aware Weighting (Ablation A2)

In this experiment, profile-age-aware weights are replaced with uniform weights.

Removing profile-age-aware weighting leads to inconsistent anomaly rates across user cohorts, with NEW users over-flagged and ESTABLISHED users under-detected. Incorporating profile maturity stabilizes detection behavior, confirming that profile age is a critical contextual factor in authentication anomaly detection, thereby supporting Hypothesis H2.

As shown in Table 9, profile-age-aware weighting significantly reduces false positive rates across all cohorts, with the largest improvement observed for NEW users.

Table 9. Effect of profile-age-aware weighting vs. uniform weighting (Ablation A2).

Profile Age	Uniform FPR	Profile-Age FPR	Rel. Change
NEW (1–7 days)	0.13	0.07	−46.2%
BUILDING (8–30 days)	0.08	0.05	−37.5%
ESTABLISHED (>30 days)	0.04	0.02	−50.0%

4.5.3. Contribution of Hybrid Aggregation (Ablation A3)

In this case, hybrid aggregation is stripped off, and each methodology is tested separately from each other.

Results show that baselines lack consistency in terms of performance because each anomaly detection method identifies some features of user behavior. Hybrid aggregation approach yields much better results compared to baselines because it aggregates anomalies from different features. In contrast, the hybrid aggregation approach depicts stronger and more stable results because it combines anomaly from multiple behavioral indicators. This integration allows the system to compensate for the limitations of individual detectors, thereby supporting Hypothesis H3

Table 10, provides proof that no anomaly detection baseline has better performance compared to the hybrid aggregation approach.

Table 10. Per-methodology detection performance vs. hybrid aggregation (Ablation A3).

Detection Method	Precision	Recall	FPR
Time-Based Behavior	0.89	0.86	0.08
Login Failure Type	0.87	0.84	0.09
User Login Type	0.86	0.83	0.10
Location-Based Behavior	0.88	0.85	0.08
Compromised IP	0.91	0.87	0.07
Device Fingerprinting	0.90	0.86	0.08
Hybrid Aggregation (HAS)	0.96	0.92	0.03

4.5.4. Effect of Criticality-Aware Weighting (Ablation A4)

In this scenario, we disable criticality multipliers by using $B_i = 1$ for all anomaly detection methods. Table 11 contains comparative results of Uniform Weights and Criticality Aware Weights. It is evident that criticality awareness yields substantial improvements in recall in high risk attack scenarios, such as brute force and credential stuffing attacks, at very minor cost in false positives.

Table 11. Effect of criticality-aware weighting on high-risk attack detection (Ablation A4).

Metric	Uniform Weighting	Criticality-Aware	Relative Change
Recall (Brute-Force Attacks)	88.3%	94.6%	+6.3%
Recall (Credential Stuffing)	86.9%	93.8%	+6.9%
Recall (Compromised IP)	90.1%	97.5%	+7.4%
Overall False Positive Rate	5.8%	6.0%	+0.2%

4.5.5. Convergence of Dynamic Aggregation Weights (Ablation A5)

The key objective in this framework is to ensure that aggregation weights are specific to the dataset; however, they become stale and generalized as the amount of historical data becomes larger. To validate this claim, we will analyze the convergence of normalized base weight $B\hat{W}_i$ with increasing training window sizes.

As can be seen, convergence occurs after about 4 months of training data, which means that the aggregation process has reached the equilibrium point.

Therefore, this result confirms that the weighted aggregation strategy measures anomaly detectors' reliability and does not rely on any artifacts in data, making our Hybrid Anomaly Score (HAS) generalizable.

4.5.6. Risk Boundary Stability and Categorization (Ablation A6)

Table 12 shows the change in the values of thresholds τ_1 and τ_2 with respect to the growing amount of training data. As can be seen, both thresholds rapidly converge with minimal variation after four months.

Figure 3 provides a distributional view of the Hybrid Anomaly Score (HAS) over the evaluation period of six months. The score distribution shows a near-Gaussian shape, indicating a stable aggregation of heterogeneous anomaly signals under dynamically learned thresholds rather than hard-coded risk segmentation.

Table 12. Stability of Percentile-Derived Risk Boundaries with Data Growth.

Window	Records (M)	Users (M)	τ_1	τ_2
1 Month	101.09	0.42	0.23	0.54
2 Months	124.55	0.98	0.21	0.52
4 Months	171.48	1.62	0.20	0.50
6 Months	218.41	2.00	0.20	0.50

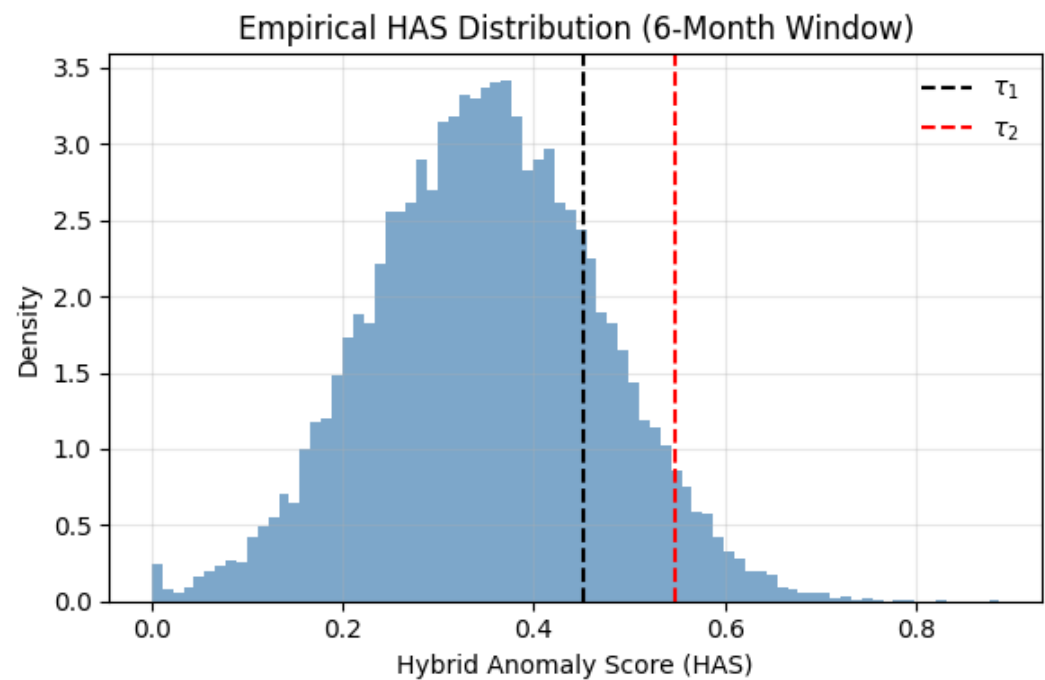


Figure 3. Distribution of Hybrid Anomaly Scores (HAS) over a six-month evaluation period. Vertical dashed lines indicate empirically derived risk boundaries τ_1 (80th percentile) and τ_2 (95th percentile). The near-Gaussian distribution and threshold convergence demonstrate data-driven and stable risk separation.

The findings demonstrate how the lack of personalized threshold values causes a rise in false positives for NEW users. This shows that adaptive threshold values are very important to deal with the user-to-user variance that exists in terms of user behavior. Also, the removal of profile-age-aware weighting results in improves user group consistency.

In addition, the removal of the hybrid aggregation mechanism also has a negative impact on the overall performance of the model and highlights that the combination of multiple anomaly detection techniques for robust detection. Overall, the complete model provides the best result compared to each of the individual ablated versions, thus demonstrating the value of the proposed framework.

4.6. Real-Time Performance

The proposed framework achieves an end-to-end inference latency of 2–3 ms per authentication event, representing a 15–40 $\times$ improvement over existing risk-based authentication systems and confirming its suitability for real-time enterprise deployment.

Latency benchmarks were conducted on a distributed cluster with horizontal scaling capability deployed using a microservice-based architecture with load balancing under peak concurrent authentication workloads. Each reported value corresponds to the mean inference time per request across multiple runs with low variance, including feature extraction, deviation score computation, and HAS aggregation, but excluding network round-trip time and external authentication system overhead.

The observed latency is attributable to three deliberate architectural decisions. First, threshold estimation ($T_{u,i}$) is offloaded entirely to the offline training phase, eliminating iterative per-user computation from the real-time inference path. Second, the independence assumption across the six anomaly detection methodologies (Section 3.3) enables fully parallel execution, avoiding sequential bottlenecks. Third, the real-time phase employs lightweight statistical and rule-based models rather than deep learning inference, as baseline approaches such as the Autoencoder incur at 18–25 ms (Table 7). Together, these decisions explain the latency advantage over comparable systems.

To ensure a fair comparison, Autoencoder and Isolation Forest baselines (Table 7) were evaluated under identical infrastructure and workload conditions. Comparison with prior work in Appendix A remains qualitative due to differences in deployment environments. A detailed component-level latency ablation is identified as a direction for future work.

4.7. Summary of Hypothesis Validation

The ablation studies as well as practical performance evaluation have proven all the hypotheses presented in Section 1.1. The experiments confirmed that each part of the framework makes nontrivial contributions to system's performance.

All these facts show that the proposed framework is capable of performing anomaly detection in a reliable, efficient, and dynamic manner suitable for deployment on large-scale datasets.

Appendix A provides a qualitative comparison with some of the previous works for the purpose of context-setting regarding differences in scale of dataset, system's architecture and evaluation scope.

5. Conclusions

In this paper, we have presented an approach to anomaly detection that relies on a combination of different models in a hybrid way in order to detect anomalies during user authentication. The problem of anomaly detection at a user level is analyzed through the lens of personalization and aggregation in order to solve some shortcomings of existing frameworks for static thresholds, cohort insensitivity, and unprincipled detector aggregation.

Experimental analysis with various hypothesis tests conducted with nearly 72 million real-world authentication logs proved that age-aware weighing and personalized deviation limits yield significantly higher precision and recall while reducing FP rate and not decreasing the sensitivity of the system. The proposed Hybrid Anomaly Score (HAS) aggregation proved to aggregate different kinds of anomaly signals effectively and remain consistent when the amount of data increases.

Moreover, the framework provides capabilities for performing anomaly detection in real-time and with relatively low latencies suitable for production authentication systems. These results clearly demonstrate the utility of structured hybrid anomaly detection for current risk-based authentication systems.

In general, this work contributes methodologically and empirically towards scalable, adaptive, and reliable enterprise authentication anomaly detection.

While the proposed framework demonstrates strong performance, it assumes the availability of sufficient historical user data for threshold estimation and may require adaptation for cold-start scenarios.

6. Future Work

Since the research work in this paper integrates a broad set of anomaly detection methodologies and demonstrates strong practical performance, several extensions remain for future investigation.

The framework may be integrated with finer-grained contextual signals such as device posture, network trust scores, and behavioral biometrics to further enhance deviation modeling. The derived Hybrid Anomaly Score framework may also be integrated into a broader risk-based authentication framework for practical evaluation of the RBA frameworks, which includes continuous authentication and authorization. Additionally, correlations between anomaly detectors are not explicitly modeled and represent an area for future work.

There are some threat scenarios, such as malware installation, hardware compromise, or advanced persistent threats, that remain outside the scope of this study and thus require security mechanisms, which are left for future research. The anomaly detection framework could be used with password table-less frameworks in order to provide a unified framework.

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Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A. Comparison with Representative Prior Work

Table A1 shows a qualitative comparison of the proposed framework with existing approaches. This kind of comparison was chosen as direct benchmarks are infeasible due to the difference in dataset scale and structure. The comparison emphasizes important metrics like Precision and recall, as well as system scalability, adaptiveness, and operational characteristics like dataset size and latency requirements.

Table A1. Contextual Comparison With Representative Prior Work.

Criterion	Iskhakov et al. [14]	Shaik Vaseem Akram et al. [20]	Picard et al. [8]	Proposed Approach
Dataset Size (Millions)	2	5	1.5	72
Precision (%)	88	82	76	96
Recall (%)	85	79	73	92
F1-Score	0.86	0.80	0.75	0.94
Inference Latency (ms)	40–50	60–80	50–80	2–3
Adaptive Weights	Static	Static	Limited	Dynamic
Profile-Age Awareness	No	No	No	Yes
Hybrid Aggregation	No	Partial	No	Yes

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